PVsyst - Simulation report

Grid-Connected System

Project: Trabajo_DataCenter

Variant: Real

Sheds, single array

System power: 102 kWp

Tres Cantos - Spain

Author

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09/01/26 19:29
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Project summary

Geographical Site
Tres Cantos
Spain

Situation

Latitude	40.60 °(N)
Longitude	-3.72 °(W)
Altitude	741 m
Time zone	UTC+1

Project settings
Albedo 0.20

Weather data
Tres Cantos
PVGIS api TMY

System summary

Grid-Connected System
Sheds, single array

Orientation #1
Fixed plane
Tilt/Azimuth 10 / -30 °

Near Shadings
Linear shadings : Slow (simul.)

User's needs
Monthly values
No grid reinjection

System information

PV Array

Nb. of modules	216 units
Pnom total	102 kWp

Inverters

Nb. of units	0.9 unit
Total power	90 kWac
Pnom ratio	1.13

Results summary

Produced Energy	166.55 MWh/year	Specific production	1641 kWh/kWp/year	Perf. Ratio PR	84.67 %
Used Energy	1259.99 MWh/year			Solar Fraction SF	13.22 %

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General parameters

Grid-Connected System

Sheds, single array

Orientation #1

Fixed plane
Tilt/Azimuth 10 / -30 °

Sheds configuration
Nb. of sheds 27 units
Single array

Shading limit angle
Limit profile angle 44.9 °

Sizes

Sheds spacing	1.42 m
Sensitive width	1.21 m
GCR Shading	86.3 %
Top inactive band	0.02 m
Bottom inactive band	0.02 m

Models used

Transposition	Perez
Diffuse	Imported
Circumsolar	separate

Horizon
Average Height 1.5 °

Near Shadings
Linear shadings : Slow (simul.)

User's needs
Monthly values
No grid reinjection

Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sep.	Oct.	Nov.	Dec.	Year	
96.6	92.4	101	98.0	109	116	121	119	109	107	97.1	95.3	1260	MWh

PV Array Characteristics

PV module

Manufacturer	Generic
Model	REC470AA Pure-RX
(Original PVsyst database)	
Unit Nom. Power	470 Wp
Number of PV modules	216 units
Nominal (STC)	102 kWp
Modules	18 string x 12 In series
At operating cond. (50°C)	
Pmpp	95.5 kWp
U mpp	619 V
I mpp	154 A

Inverter

Manufacturer	Generic
Model	SUN2000-100KTL-M2-400Vac
(Original PVsyst database)	
Unit Nom. Power	100 kWac
Number of inverters	9 * MPPT 10% 0.9 unit
Total power	90.0 kWac
Operating voltage	200-1000 V
Max. power (=>40°C)	110 kWac
Pnom ratio (DC:AC)	1.13
No power sharing between MPPTs	

Total PV power

Nominal (STC)	102 kWp
Total	216 modules
Module area	450 m²
Cell area	419 m²

Total inverter power

Total power	90 kWac
Nb. of inverters	1 unit
	0.1 unused
Pnom ratio	1.13

Array losses

Array Soiling Losses
Loss Fraction 1.0 %

Thermal Loss factor
Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses
Global array res. 64 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss
Loss Fraction -0.53 %

Module mismatch losses
Loss Fraction 2.00 % at MPP

Strings Mismatch loss
Loss Fraction 0.05 %

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Array losses

IAM loss factor

Incidence effect (IAM): User defined profile

0°	30°	45°	60°	70°	75°	80°	85°	90°
1.000	1.000	1.000	0.974	0.907	0.832	0.688	0.445	0.000

System losses

Unavailability of the system

Time fraction 0.5 %
1.8 days,
3 periods

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Horizon definition

Horizon from PVGIS website API, Lat=40°35'53", Long=-3°43'2", Alt=741m

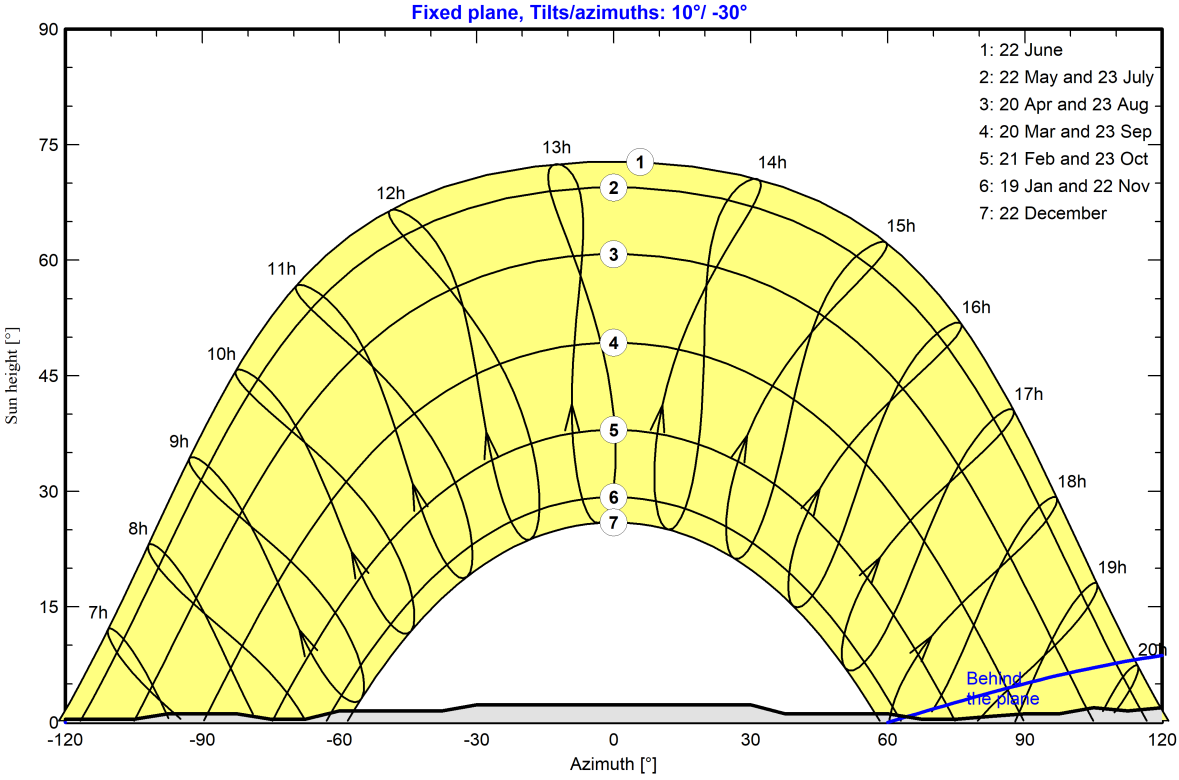
Average Height	1.5 °	Albedo Factor	0.91
Diffuse Factor	1.00	Albedo Fraction	100 %

Horizon profile

Azimuth [°]	-180	-173	-158	-150	-143	-105	-98	-83	-75	-68	-60
Height [°]	1.9	1.5	0.8	0.8	0.4	0.4	1.1	1.1	0.4	0.4	1.5
Azimuth [°]	-38	-30	30	38	60	68	75	83	90	98	105
Height [°]	1.5	2.3	2.3	1.1	1.1	0.4	0.4	0.8	1.1	1.1	1.9
Azimuth [°]	113	120	128	135	143	150	158	165	173	180	
Height [°]	1.5	1.9	2.3	2.7	2.7	3.1	2.7	1.9	1.5	1.9	

Sun Paths (Height / Azimuth diagram)

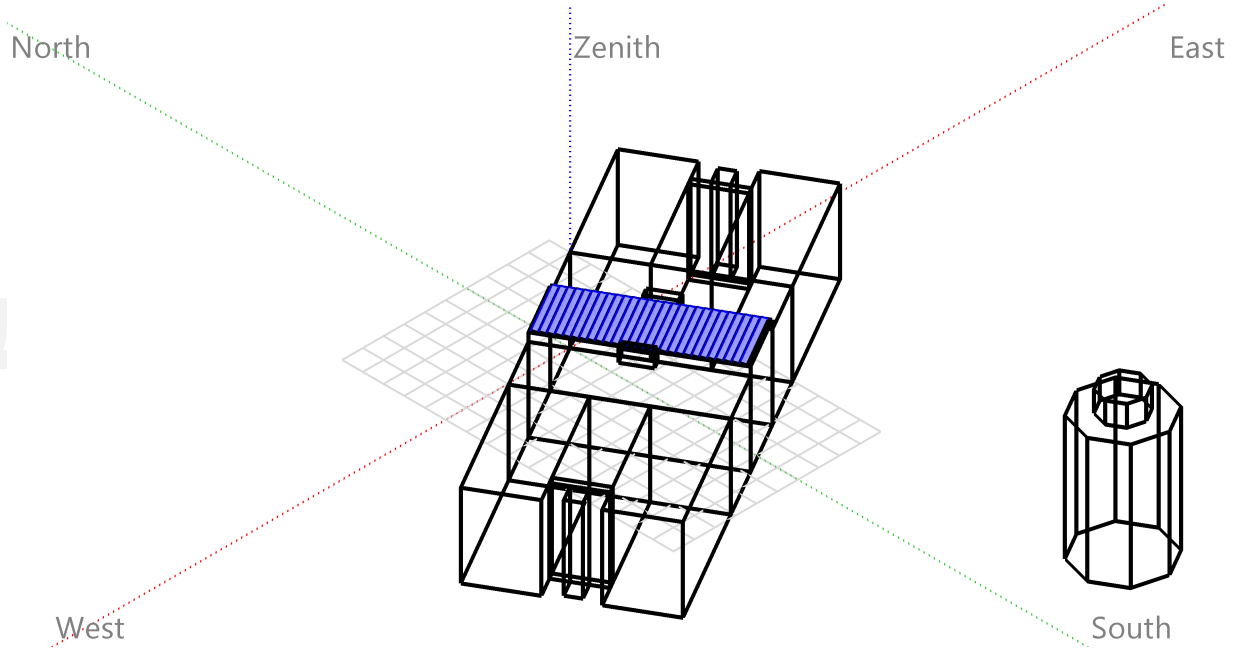
Orientation #1



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Near shadings parameter

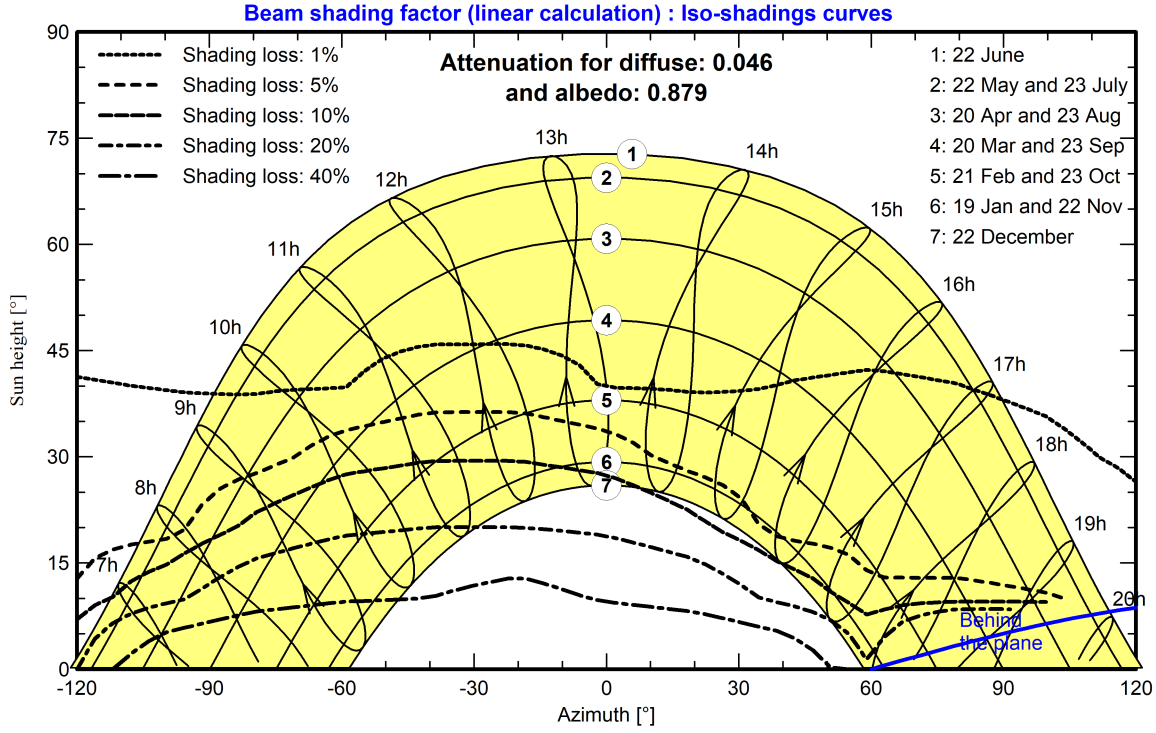
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 10°/ -30°

Beam shading factor (linear calculation) : Iso-shadings curves



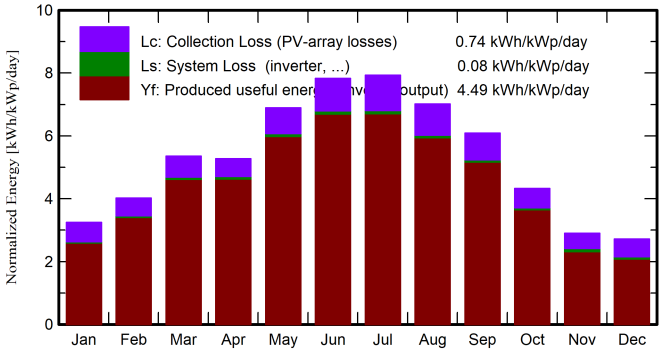
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Main results

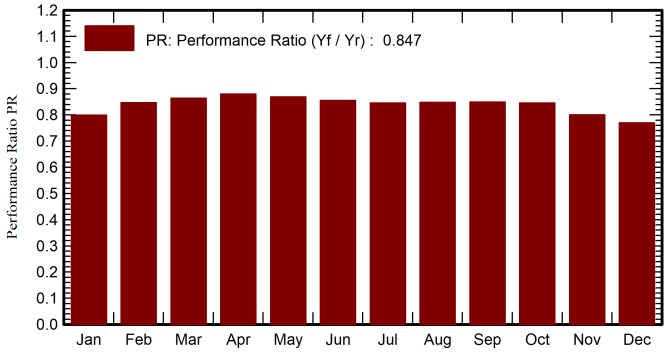
System Production

Produced Energy	166.55 MWh/year	Specific production	1641 kWh/kWp/year
Used Energy	1259.99 MWh/year	Perf. Ratio PR	84.67 %
		Solar Fraction SF	13.22 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

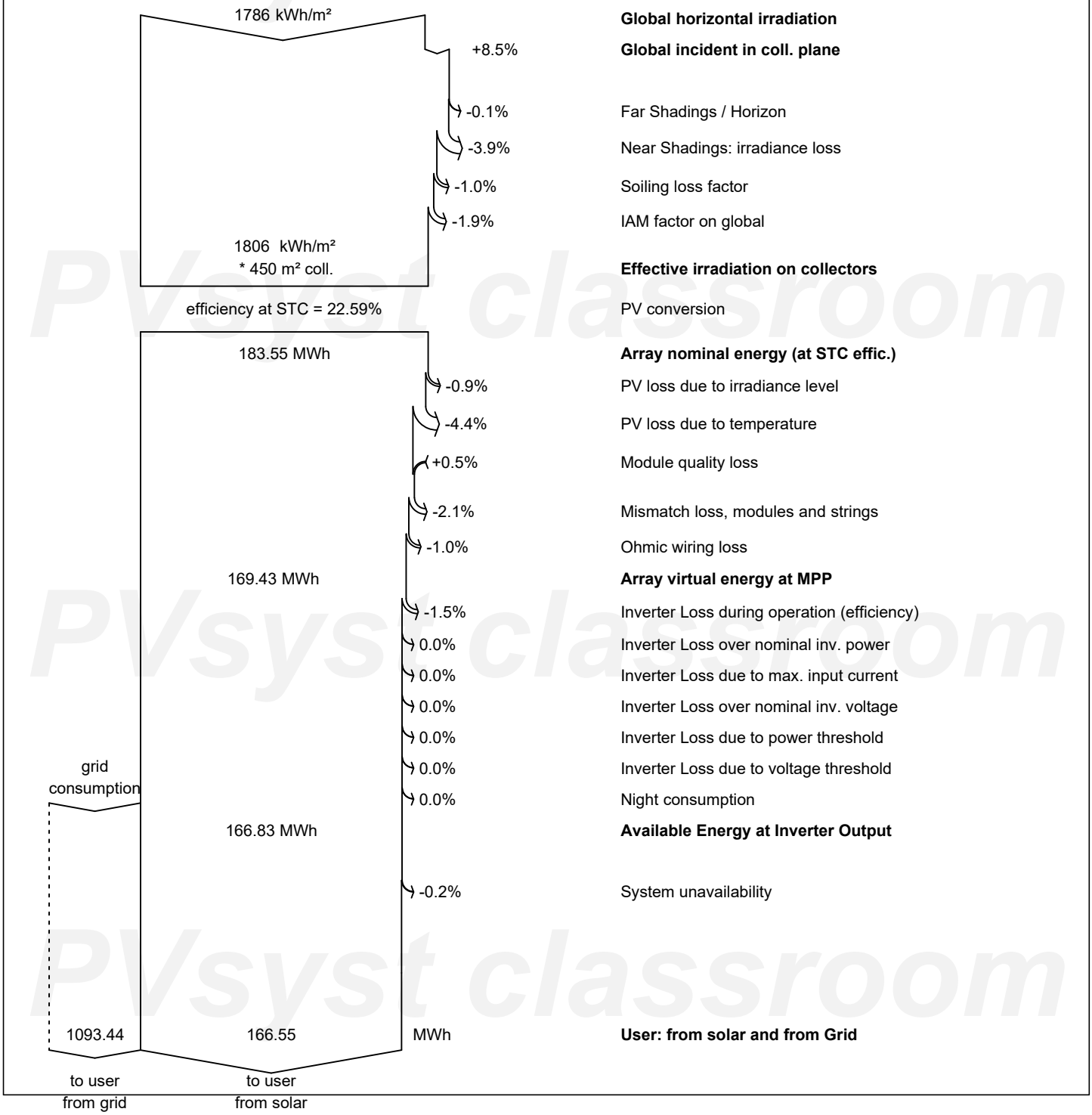
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray MWh	E_User MWh	E_Solar MWh	EUnused MWh	EFrGrid MWh
January	79.2	22.95	3.48	100.7	84.1	8.3	96.6	8.2	0.00	88.4
February	95.3	27.77	6.15	112.7	101.1	9.8	92.4	9.7	0.00	82.7
March	149.8	45.66	10.02	166.1	155.0	14.8	101.0	14.6	0.00	86.5
April	151.5	68.96	9.79	158.1	149.7	14.3	98.0	14.1	0.00	83.9
May	207.5	76.51	17.21	213.8	204.9	19.1	108.8	18.9	0.00	89.9
June	232.2	72.53	23.55	235.0	226.2	20.7	115.5	20.4	0.00	95.1
July	240.9	60.21	27.55	245.8	236.8	21.4	121.4	21.1	0.00	100.3
August	207.4	49.78	25.85	217.4	208.5	19.0	118.8	18.7	0.00	100.1
September	167.1	46.02	21.91	182.5	172.9	16.0	108.5	15.7	0.00	92.8
October	116.6	37.82	12.80	134.2	122.5	11.7	106.5	11.5	0.00	95.0
November	72.4	28.28	7.02	87.0	75.7	7.4	97.1	7.1	0.00	90.0
December	65.9	21.96	4.82	84.2	69.0	6.8	95.3	6.6	0.00	88.7
Year	1785.8	558.44	14.23	1937.6	1806.3	169.4	1260.0	166.5	-0.01	1093.4

Legends

GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_User	Energy supplied to the user
T_Amb	Ambient Temperature	E_Solar	Energy from the sun
GlobInc	Global incident in coll. plane	EUnused	Unused energy (no grid injection)
GlobEff	Effective Global, corr. for IAM and shadings	EFrGrid	Energy from the grid

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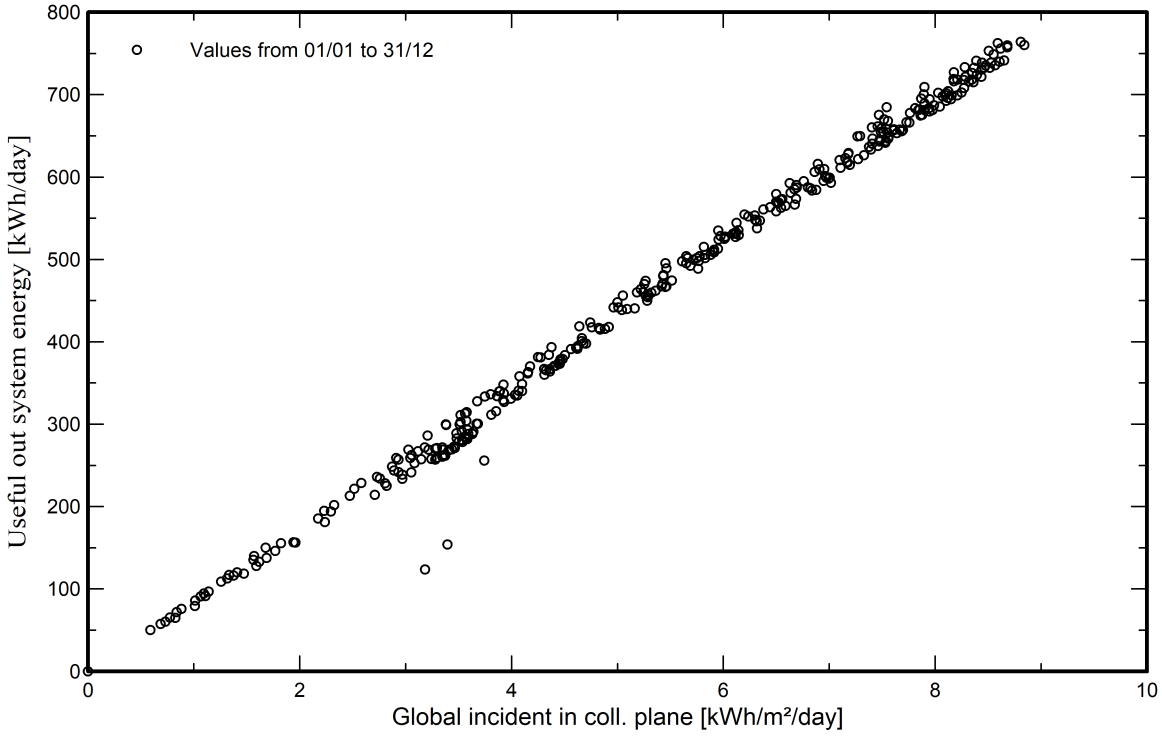
Loss diagram



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Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

